

星际介质天文学作业

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18.1 由 (18.9) 式:

$$\frac{I([\text{O III}]5008)}{I(\text{H}\beta)} = \frac{n(\text{O III})}{n(\text{H}^+)} \times \frac{k_{03}E_{32}A_{32}}{\alpha_{\text{eff,H}\beta}E_{\text{H}\beta}(A_{31} + A_{32})}$$

可得:

$$\frac{n(\text{O III})}{n(\text{H}^+)} = \frac{I([\text{O III}]5008)}{I(\text{H}\beta)} \times \frac{\alpha_{\text{eff,H}\beta}E_{\text{H}\beta}(A_{31} + A_{32})}{k_{03}E_{32}A_{32}}$$

通过查表可以得知:

$$A_{31} = 6.215 \times 10^{-3}, \quad A_{32} = 1.814 \times 10^{-2}$$

$$E_{32} = \frac{hc}{5008.2 \text{ \AA}}, \quad E_{\text{H}\beta} = \frac{hc}{4862.7 \text{ \AA}}$$

则有:

$$\begin{aligned} \frac{n(\text{O III})}{n(\text{H}^+)} &= \frac{I([\text{O III}]5008)}{I(\text{H}\beta)} \times \frac{\alpha_{\text{eff,H}\beta}}{k_{03}} \times \frac{5008.2 \text{ \AA}}{4862.7 \text{ \AA}} \times \frac{6.215 \times 10^{-3} + 1.814 \times 10^{-2}}{1.814 \times 10^{-2}} \\ &= \frac{I([\text{O III}]5008)}{I(\text{H}\beta)} \times \frac{\alpha_{\text{eff,H}\beta}}{k_{03}} \times 1.3828 \end{aligned}$$

由 (14.9) 式得:

$$\alpha_{\text{eff,H}\beta} = 3.03 \times 10^{-14} T_4^{-0.874-0.058 \ln T_4} \text{ cm}^3/\text{s} \approx 3.03 \times 10^{-14} T_4^{-0.87} \text{ cm}^3/\text{s}$$

又由 (18.10) 式得:

$$k_{03} = 8.629 \times 10^{-8} T_4^{-1/2} \frac{\Omega_{03}}{g_0} e^{-E_{30}/kT} \text{ cm}^3/\text{s}$$

其中 $E_{30}/k = 29170 \text{ K}$, $g_0 = 1$, 且由表 (F.2) 可得:

$$\Omega_{03} = 0.243 \times T_4^{0.120+0.031 \ln T_4}$$

可以近似 Ω_{03} 与温度无关, 即:

$$\Omega_{03} \approx 0.243$$

于是有:

$$\begin{aligned} k_{03} &= 8.629 \times 10^{-8} T_4^{-1/2} \frac{\Omega_{03}}{g_0} e^{-E_{30}/kT} \text{ cm}^3/\text{s} \\ &= 8.629 \times 10^{-8} T_4^{-0.5} \times 0.243 e^{-29170 \text{ K}/T} \text{ cm}^3/\text{s} \\ &= 2.097 \times 10^{-8} T_4^{-0.5} e^{-2.917/T_4} \text{ cm}^3/\text{s} \end{aligned}$$

则有：

$$\begin{aligned}\frac{\alpha_{\text{eff,H}\beta}}{k_{03}} &= \frac{3.03 \times 10^{-14} T_4^{-0.874}}{2.097 \times 10^{-8} T_4^{-0.5} e^{-2.917/T_4}} \\ &= 1.445 \times 10^{-6} T_4^{-0.37} e^{-2.917/T_4}\end{aligned}$$

因此：

$$\begin{aligned}\frac{n(\text{O III})}{n(\text{H}^+)} &= \frac{I([\text{O III}]5008)}{I(\text{H}\beta)} \times \frac{\alpha_{\text{eff,H}\beta}}{k_{03}} \times 1.3828 \\ &= \frac{I([\text{O III}]5008)}{I(\text{H}\beta)} \times (1.445 \times 10^{-6} T_4^{-0.37} e^{-2.917/T_4}) \times 1.3828 \\ &= 2.00 \times \frac{I([\text{O III}]5008)}{I(\text{H}\beta)} T_4^{-0.37} e^{-2.917/T_4}\end{aligned}$$

即常数 $C = 2.00$ 。

18.2 (a) 由图 (18.2) 和图 (18.4) 可知：

$$\frac{I([\text{O III}]4364.4 \text{ \AA})}{I([\text{O III}]5008.2 \text{ \AA})} = 0.003 \Rightarrow \begin{cases} T = 7365.18 \text{ K}, & n_e = 10^5 \text{ cm}^{-3} \\ T = 7971.36 \text{ K}, & n_e = 10^4 \text{ cm}^{-3} \\ T = 8055.57 \text{ K}, & n_e \leq 10^3 \text{ cm}^{-3} \end{cases}$$

$$\frac{I([\text{O II}]3729.8 \text{ \AA})}{I([\text{O II}]3727.1 \text{ \AA})} = 1.2 \Rightarrow n_e T_4^{-1/2} = 241.524 \text{ cm}^{-3}$$

可见 $T \approx 8000 \text{ K}$ ，即 $T_4^{-1/2} \approx 0.8^{-1/2} = 1.12$ ，而 $n_e T_4^{-1/2} = 241.524$ 说明 n_e 显然小于 10^3 cm^{-3} 。

取电子温度 $T = 8055.57 \text{ K}$ ，得电子数密度：

$$n_e = \frac{241.524}{0.805557^{-1/2}} = 216.775 \text{ cm}^{-3}$$

综上，估计电子温度和数密度为 $T = 8.06 \times 10^3 \text{ K}$ ， $n_e = 217 \text{ cm}^{-3}$ 。

18.2 (b) 由视星等公式：

$$m - m_0 = -2.5 \log\left(\frac{I}{I_0}\right) + C$$

可得修正后 O III 的谱线强度比为：

$$\frac{I([\text{O III}]4364.4 \text{ \AA})}{I([\text{O III}]5008.2 \text{ \AA})} = \frac{0.003}{10^{-0.4/2.5}} = 0.004$$

则由图 (18.2) 可得电子温度为：（仍然假设 $n_e \leq 10^3 \text{ cm}^{-3}$ ）

$$T = 8641.97 \text{ K}$$

得电子数密度：

$$n_e = \frac{241.524}{0.864197^{-1/2}} = 224.526 \text{ cm}^{-3}$$

综上，估计电子温度和数密度为 $T = 8.64 \times 10^3 \text{ K}$ ， $n_e = 225 \text{ cm}^{-3}$